Term values in the term model

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the lemma sits inside the FOL tag branch and is delivered as the author wrote it

Editorial note. The label lem:val-in-termmodel names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Term model definition. Same unit, \ollabel{defn:termmodel} (source line 1186). The interpretation of constants and function symbols in M(Gamma*) that the induction reads off.

Source reference. Original definition defn-termmodel: sources/olp-first-order-logic-completeness/pinned-original-source.tex, source lines 1186–1186. Prerequisite pointer invoked at proof line(s) 1228; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let* $\mathfrak{M}(\Gamma^*)$ *be the term model of defn-termmodel, then* $\mathrm{val}(t)M(\Gamma^*) = t$.

Proof. The proof is by induction on t, where the base case, when t is constant, follows directly from the definition of the term model. For the induction step assume $t_1, \ldots, t_n$ are closed terms such that $\mathrm{val}(t_i)M(\Gamma^*) = t_i$ and that f is an n-ary function. Then

$$\begin{aligned}\mathrm{val}(f(t_1, \ldots, t_n))M(\Gamma^*) &= sfM(\Gamma^*)(\mathrm{val}(t_1)M(\Gamma^*), \ldots, \mathrm{val}(t_n)M(\Gamma^*)) \\ &= sfM(\Gamma^*)(t_1, \ldots, t_n) \\ &= f(t_1, \ldots, t_n),\end{aligned}$$

and so by induction this holds for every closed term t. □